

Cash Flow Analysis

Compound interest factors for Discrete Compounding

- Because of the difficulties of making precise calculations of complex and irregular cash flows, engineers usually work with fairly **simple models of cash flow patterns**. The most common type of model assumes that all cash flows and all compounding of cash flows occur at the ends of conventionally defined periods, such as months or years. Models that make this assumption are called **discrete models**.

The four discrete cash flow patterns are:

1. A single disbursement (money spent) or receipt (money received)
2. A set of equal disbursements or receipts over a sequence of periods, referred to as an **annuity**
3. A set of disbursements or receipts that change by a constant *amount* from one period to the next in a sequence of periods, referred to as an **arithmetic gradient series**
4. A set of disbursements or receipts that change by a constant *proportion* from one period to the next in a sequence of periods, referred to as a **geometric gradient series**

Compound interest factors are formulas that define mathematical equivalence for specific common cash flow patterns.

Single Disbursement or receipt

- In many situations, a single disbursement or receipt is an appropriate model of cash flows. For example, the salvage value of production equipment with a limited service life will be a single receipt at some future date. An investment today to be redeemed at some future date is another example.
- Two commonly used factors relate a single cash flow in one period to another single cash flow in a different period. They are the ***compound amount factor*** and the ***present worth factor***.
- The **compound amount factor**, denoted by $(F/P, i, N)$, gives the future amount, F , that is equivalent to a present amount, P , when the interest rate is i and the number of $F = P(1 + i)^N$
- Compound interest Equation:
- Compound interest Factor: $F = P(1 + i)^N = P(F/P, i, N)$

Single Disbursement or receipt

How much money will be in a bank account at the end of 15 years if \$100 is invested today and the nominal interest rate is 8 percent compounded semi-annually?

$F = ?$, $n = 15$ years, $P = \$100$, $r = 0.08/$ year, $m = 2$, $N = 30$,

Using Interest Equations:

$$F = P (1 + i)^N$$

$$F = \underline{\hspace{2cm}}$$



Using Interest Factors:

$$F = P (F/P, i, N)$$

$$F = 100 (F/P, 4\%, 30)$$

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Using Interest Equations:

$$F = P (1 + i)^N$$

$i = r/m$

$$F = 100 (1 + 0.08/2)^{30}$$

$N = n*m$

$$F = \$ 324.34$$

Using Interest Factors:

$$F = P (F/P, i, N)$$

$$F = 100 (F/P, 4\%, 30)$$

go to page with 4% mentioned on top- then to the row $n=30$, then multiply the f/p with P

$$F = 100 (3.2434)$$

$$F = \$ 324.34$$



Compound interest factors for Single Disbursement or receipt

- **The Single Payment present worth factor**, denoted by $(P/F, i, N)$, gives the present amount, P , that is equivalent to a future amount, F , when the interest rate is i and the number of periods is N . The present worth factor is the inverse of the compound amount factor, $(F/P, i, N)$. It gives the present worth, P , of a future amount, F .

$$(P/F, i, N) = \frac{1}{(1 + i)^N}$$

Suppose that \$1,000 is to be received in five years. At an annual interest rate of 12%, what is the present worth of this amount?

SOLUTION

Given: $F = \$1,000$, $i = 12\%$ per year, and $N = 5$ years.

Find: P .

Compound interest factors for Single Disbursement or receipt

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SOLUTION

Given: $F = \$1,000$, $i = 12\%$ per year, and $N = 5$ years.

Find: P .

$$P = \$1,000 \overbrace{(P/F, 12\%, 5)}^{(0.5674)} = \$567.40.$$

$$P = \$1,000(1 + 0.12)^{-5} = \$1,000(0.5674) = \$567.40.$$



Compound interest factors for **Equal Payment Series or Annuities**

A **series of uniform receipts or disbursements** that **start at the end of the first period and continue over N periods** is called an annuity. **Mortgage or lease payments** and maintenance contract fees are examples of the annuity cash flow pattern.

You financed a house purchasing plan on bank lease. For next 10 years at 10 percent rate. You have to pay fixed amount to the bank at end of year for 10 years.

Uniform Series Future Worth Factor & Sinking Fund Factor

- The **sinking fund factor**, denoted by $(A/F, i, N)$, gives the size, A , of a repeated receipt or disbursement that is equivalent to a future amount, F , if the interest rate is i and the number of periods is N . The name of the factor comes from the term **sinking fund**. A sinking fund is an interest-bearing account into which regular deposits are made in order to accumulate some amount.
- The sinking fund factor is commonly used to determine how much has to be set aside or saved per period to accumulate an amount F at the end of N periods at an interest rate i . The amount F might be used, for example, to purchase new or replacement equipment, to pay for renovations, or to cover capital expenditures. $(A/F, i, N) = \frac{i}{(1 + i)^N - 1}$

The Kelowna Go-Kart Klub has decided to build a clubhouse and track five years from now. It must accumulate \$50 000 by the end of five years by setting aside a uniform amount from its dues at the end of each year. If the interest rate is 10 percent, how much must be set aside each year?

$F = 50000$, $i = 10\%$, $n = 5$ yrs., $A = ?$

$$(A/F, i, N) = \frac{i}{(1 + i)^N - 1}$$

$A =$

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$$(A/F, i, N) = \frac{i}{(1 + i)^N - 1}$$

F = 50000, i = 10%, n = 5 yrs., A = ?

$$A = F (A/F, i, N)$$

$$A = 50000 * (0.01 / [(1.01)^5 - 1])$$

$$A = \underline{\hspace{2cm}}$$

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$$(A/F, i, N) = \frac{i}{(1 + i)^N - 1}$$

F = 50000, i = 10%, n = 5 yrs., A = ?

yani future kay lye paise chahiye that is 50000
har saal apko paise hata kar rakhne hai which is annuity

$$A = F (A/F, i, N)$$

$$A = 50000 * (0.01 / [(1.01)^5 - 1])$$

$$A = 8190.00$$



$$\begin{aligned} A &= 50\,000(A/F, 10\%, 5) \\ &= 50\,000(0.1638) \\ &= 8190.00 \end{aligned}$$

uniform series compound amount factor /USFWF

- The **uniform series compound amount factor**, denoted by $(F/A, i, N)$, gives the future value, F , that is equivalent to a series of equal-sized receipts or disbursements, A , when the interest rate is i and the number of periods is N .

$$(F/A, i, N) = \frac{(1 + i)^N - 1}{i}$$

Suppose you make an annual contribution of \$3,000 to your savings account at the end of each year for 10 years. If the account earns 7% interest annually, how much can be withdrawn at the end of 10 years (Figure 3.19)?

Given: $A = \$3,000$, $N = 10$ years, and $i = 7\%$ per year.
Find: F .

$$F = A \left[\frac{(1 + i)^N - 1}{i} \right] = A(F/A, i, N).$$

$$F = 3000 (F/A, 0.07, 10)$$

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$$F = 3000 (F/A, 0.07, 10)$$

$$F = 3000 (13.8164)$$

$$F = 41, 449.$$



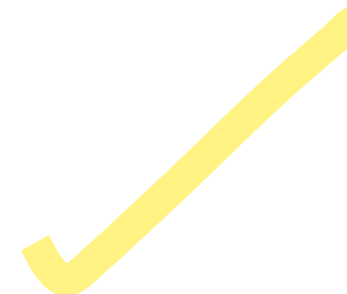
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$$F = A \left[\frac{(1 + i)^N - 1}{i} \right] = A(F/A, i, N).$$

$$F = 3000 [((1+0.07)^{10} - 1) / 0.07]$$

$$F = 41,449.20$$



Capital recovery factor Vs. Uniform Series Present Worth Factor

- The **capital recovery factor**, denoted by $(A/P, i, N)$, gives the value, A , of the equal periodic payments or receipts that are equivalent to a present amount, P , when the interest rate is i and the number of periods is N .
- The objective of this mode of investment is to find the annual equivalent amount (A) which is to be recovered at the end of every interest period for n interest periods for a loan (P) which is sanctioned now at an interest rate of I compounded at the end of every interest period. Or to find out, for example, how much money must be saved over N future periods to “recover” a capital investment of P today. Think of Lease Paym

$$A = P \left[\frac{i(1 + i)^N}{(1 + i)^N - 1} \right] = P(A/P, i, N).$$

EXAMPLE 3.6 A bank gives a loan to a company to purchase an equipment worth Rs. 10,00,000 at an interest rate of 18% compounded annually. This amount should be repaid in 15 yearly equal installments. Find the installment amount that the company has to pay to the bank.

Solution

$$P = \text{Rs. } 10,00,000$$

$$i = 18\%$$

$$n = 15 \text{ years}$$

$$A = ?$$

$$A = P \frac{i(1+i)^n}{(1+i)^n - 1} = P(A/P, i, n)$$

EXAMPLE 3.6 A bank gives a loan to a company to purchase an equipment worth Rs. 10,00,000 at an interest rate of 18% compounded annually. This amount should be repaid in 15 yearly equal installments. Find the installment amount that the company has to pay to the bank.

Solution

$$P = \text{Rs. } 10,00,000$$

$$i = 18\%$$

$$n = 15 \text{ years}$$

$$A = ?$$

$$A = P \frac{i(1+i)^n}{(1+i)^n - 1} = P(A/P, i, n)$$

$$= 10,00,000 \times (A/P, 18\%, 15)$$

$$= 10,00,000 \times (0.1964)$$

$$= \text{Rs. } 1,96,400$$



The annual equivalent installment to be paid by the company to the bank is Rs. 1,96,400.

Capital recovery factor Vs. Uniform Series Present Worth Factor

- The **UNIFORM series present worth factor**, denoted by $(P/A, i, N)$, gives the present amount, P , that is equivalent to an annuity with disbursements or receipts in the amount, A , where the interest rate is i and the number of periods is N . It is the reciprocal of the capital recovery factor:

$$P = A \left[\frac{(1 + i)^N - 1}{i(1 + i)^N} \right] = A(P/A, i, N).$$



Uniform Series Present Worth Factor

EXAMPLE 3.5 A company wants to set up a reserve which will help the company to have an annual equivalent amount of Rs. 10,00,000 for the next 20 years towards its employees welfare measures. The reserve is assumed to grow at the rate of 15% annually. Find the single-payment that must be made now as the reserve amount.

Solution

$$P = A \left[\frac{(1 + i)^N - 1}{i(1 + i)^N} \right] = A(P/A, i, N).$$

$$A = \text{Rs. } 10,00,000$$

$$i = 15\%$$

$$n = 20 \text{ years}$$

$$P = ?$$

EXAMPLE 3.5 A company wants to set up a reserve which will help the company to have an annual equivalent amount of Rs. 10,00,000 for the next 20 years towards its employees welfare measures. The reserve is assumed to grow at the rate of 15% annually. Find the single-payment that must be made now as the reserve amount.

Solution

$$A = \text{Rs. } 10,00,000$$

$$i = 15\%$$

$$n = 20 \text{ years}$$

$$P = ?$$

$$P = A \frac{(1 + i)^n - 1}{i(1 + i)^n} = A(P/A, i, n)$$

$$= 10,00,000 \times (P/A, 15\%, 20)$$

$$= 10,00,000 \times 6.2593$$

$$= \text{Rs. } 62,59,300$$

